Logarithms

$\Rightarrow \log_a x \Rightarrow$ Log of x with base a

$2^{\textcircled{4}} \Rightarrow 16$ → Exponents / power / index

$\Rightarrow \log_2 16 \Leftrightarrow 4$

$\Rightarrow$ THINK of it like Log base {2} should be multiplied by what number to get 16.

$\Rightarrow$ log inverse of Exponential function

$\log_2 x \Leftrightarrow y$ (input, output / output, input)

$2^y \Rightarrow x$ (input, output / output, input)

Ex) $\log_3 81 \Rightarrow 4$

$\Rightarrow \log_3 81 \Rightarrow ?$

$3^k \Rightarrow 81$

$3^k \Rightarrow 3^4$

$k \Rightarrow 4$

Ex) $\log_{27} 81 \Rightarrow ?$

$\rightarrow \log_{27} 81 \Rightarrow ?$

$27^k \Leftrightarrow 81$

$3^{3k} \Leftrightarrow 3^4$

$3k \Leftrightarrow 4$

$k \Rightarrow \frac{4}{3}$

$\log_{27} 81 \Rightarrow \frac{4}{3}$

Ex) $\log_8 32 \Rightarrow ?$

$\Rightarrow \log_8 32 \Rightarrow ?$

$\rightarrow 8^x \Rightarrow 32$

$\rightarrow (2^3)^x \Leftrightarrow 2^5$

$\rightarrow 2^{3x} \Leftrightarrow 2^5$

$\rightarrow x \Leftrightarrow \frac{5}{3}$

$\log_8 32 \Leftrightarrow \frac{5}{3}$

Ex5) $\log_{27} 9\sqrt{3} \Leftrightarrow ?$

$\rightarrow \log_{27} 9\sqrt{3} \Rightarrow ?$

$27^k \Rightarrow 9\sqrt{3}$

$(3^3)^k \Rightarrow 3^2 3^{1/2}$

$3k \Rightarrow 2 + \frac{1}{2}$

$3k \Rightarrow \frac{5}{2}$

$k \Rightarrow \frac{5}{6}$

$\log_{27} 9\sqrt{3} \Leftrightarrow \frac{5}{6}$

Ex) $\log_{\frac{1}{3}} \frac{1}{9} \Rightarrow ?$

$\Rightarrow \log_{\frac{1}{3}} \frac{1}{9} \Rightarrow ?$

$\left(\frac{1}{3}\right)^x \Rightarrow \left(\frac{1}{9}\right)$

$(3)^{-x} \Rightarrow (9)^{-1}$

$(3)^{-x} \Rightarrow (3)^{-2}$

$-x \Leftrightarrow -2$

$x \Rightarrow 2$

$\log_{\frac{1}{3}} \frac{1}{9} \Rightarrow 2$

Ex) $\log_{\frac{1}{11}} 1331 \Rightarrow k$

$\rightarrow \log_{\frac{1}{11}} 1331 \Rightarrow ?$

$\rightarrow \left(\frac{1}{11}\right)^k \Rightarrow 1331$

$\rightarrow (11)^{-k} (=) (11)^3$

$\rightarrow -k (=) 3$

$\rightarrow k \Rightarrow -3$

$\log_{\frac{1}{11}} 1331 \Rightarrow -3$

Ex) $\log_{2\sqrt{3}} 1728 \Rightarrow ?$

$\log_{2\sqrt{3}} 1728 \Rightarrow ?$

1728	$\log_{2\sqrt{3}} 1728 \Rightarrow ?$
864	
432	$(2\sqrt{3})^k \Rightarrow 1728$
216	$(2\sqrt{3})^k (=) 2^6 3^3$
108	$(2\sqrt{3})^k (=) 2^6 \sqrt{3}^6$
54	$(2\sqrt{3})^k (=) (2\sqrt{3})^6$
27	$k (=) 6$
9	
3	$\log_{2\sqrt{3}} 1728 \Rightarrow 6$
1	

Rough

Ex) $\log_2 \cos 45°$

$\Rightarrow \log_2 \cos 45°$

$\Rightarrow \log_2 1/\sqrt{2} \Rightarrow k$

$(2)^k \Rightarrow \frac{1}{\sqrt{2}}$

$(2^k) \Rightarrow \frac{1}{(2)^{1/2}}$

$(2)^k \Rightarrow (2)^{-1/2}$

$k \Rightarrow -\frac{1}{2}$

$\log_2 1/\sqrt{2} \Rightarrow -\frac{1}{2}$

Ex) $\log_{2-\sqrt{3}} 2+\sqrt{3} \Rightarrow ?$

$\Rightarrow \log_{2-\sqrt{3}} 2+\sqrt{3} \Rightarrow k$

$(2-\sqrt{3})^k \Rightarrow 2+\sqrt{3}$

Note: $(2+\sqrt{3})(2-\sqrt{3}) \Rightarrow 1$

$(a+b)(a-b) \Rightarrow 1$

$a^2 - b^2 \Rightarrow 1$

$\Rightarrow 2+\sqrt{3} \Rightarrow \frac{1}{2-\sqrt{3}}$

$(2-\sqrt{3})^k \Rightarrow \frac{1}{2-\sqrt{3}}$ {Replace}

$(2-\sqrt{3})^k \Rightarrow (2-\sqrt{3})^{-1}$

$k \Rightarrow -1$

a) $\log_{0.125} 8$

$\Rightarrow \log_{0.125} 8 \Rightarrow ?$

$\log_{.125} 8 \Rightarrow k$

$\left(\frac{125}{1000}\right)^k = 8$

$\left(\frac{5}{\cancel{10}\cancel{0}2}\right)^{3k} \Rightarrow (2)^3$

$\left(\frac{1}{2}\right)^{+3k} \Rightarrow (2)^3$

$(2)^{-3k} \Rightarrow (2)^3$

$-3k \Rightarrow 3$

$k \Rightarrow -1$

b) $\log_{\frac{3}{4}} 1.\overline{3} \Rightarrow ?$

$\log_{\frac{3}{4}} 1.\overline{3} \Rightarrow k$

$\left(\frac{3}{4}\right)^k \Rightarrow 1.\overline{3}$

$x \Rightarrow 1.33$ {Assume}

multiply by 10

$10x \Rightarrow 13.333$

$-\ x \Rightarrow 1.333$

$9x \Leftrightarrow 12.$

$x \Leftrightarrow \frac{12}{9}$

$x \Rightarrow \frac{4}{3}$

$\left(\frac{3}{4}\right)^k \Rightarrow \frac{4}{3}$

$\left(\frac{3}{4}\right)^k \Rightarrow \left(\frac{3}{4}\right)$

$k \Leftrightarrow$

Remember

Here We Wann
out the decimal
the Eq^n that's
assume it to

Ex) $\log_{2.25} 0.\overline{4} \Rightarrow ?$

$\rightarrow \log_{2.25} 0.\overline{4} \Leftrightarrow k$

$\rightarrow (2.25)^k \Rightarrow 0.\overline{4}$

$0.\overline{4} \Rightarrow x$ {assume}

$x \Rightarrow 0.4444$

multiply by 10

$10x \Rightarrow 4.444$

$-x \Rightarrow 0.444$

$9x \Rightarrow 4.0$

$x \Rightarrow \frac{4}{9}$

$\Rightarrow (2.25)^k \Leftrightarrow \left(\frac{4}{9}\right)$

$\left(\frac{225}{100}\right)^k \Rightarrow \frac{4}{9}$

$\rightarrow \left(\frac{\cancel{45}9}{\cancel{225}/\cancel{100}}\right)^k \Rightarrow \frac{4}{9}$

$\rightarrow \left(\frac{9}{4}\right)^k \Rightarrow \frac{4}{9}$

$\rightarrow \left(\frac{4}{9}\right)^{k(-1)} \Rightarrow \left(\frac{4}{9}\right)^{-1}$

$-k \Rightarrow 1$

$\rightarrow k \Rightarrow -1$

Ex) $\log_{10} 0.\overline{9} \Rightarrow ?$

$\Rightarrow \log_{10} 0.\overline{9} \Rightarrow k$

$(10)^k \Rightarrow 0.\overline{9}$

$0.999 \Rightarrow x$

$x \Rightarrow 0.999$

$10 \times x \Rightarrow 9.999$

$\rightarrow 10x \Rightarrow 9.99$

$(-)\ x \Rightarrow 0.99$

$9x \Leftrightarrow 9.$

$x \Rightarrow 1$

$(10)^k \Rightarrow 1$

$(10)^k \Leftrightarrow (10)^0$

$k \Leftrightarrow 0$

$\log_{10} 0.\overline{9} \Rightarrow 0$

Notes :-

1) $\log_a 1 \Rightarrow 0$

2) $\log_a 1/a \Rightarrow -1$

3) $\log_a a \Rightarrow 1$

Ex) $\log_{3\sqrt{7}} 2401$

$\Rightarrow \log_{3\sqrt{7}} 2401 \Rightarrow k$

$\rightarrow (7^{1/3})^k \Rightarrow 2401$

$\rightarrow (7)^{\frac{k}{3}} \Leftrightarrow (7)^4$

$\frac{k}{3} \Rightarrow 4$

$\rightarrow k \Rightarrow 12$

$\Rightarrow \log_{3\sqrt{7}} 2401 \Leftrightarrow 12$

7 | 2401	7 | 2401, 343
7 | 343	2 |
7 | 49	30
7 | 7	28 ↓
1	21

Ex) $\log_2 \log_2 \log_3 \log_3 27^3$

$\rightarrow \log_2(\log_2(\log_3(\log_3 27^3)))$

$\rightarrow \log_3 27 \Rightarrow k$

$(3)^k \Rightarrow 27^3 \Rightarrow (3)^{3\times3}$

$k = 9$

$\rightarrow \log_2 \log_2(\log_2 9)$

$\rightarrow \log_2 \log_2(\log_3 9)$

$\rightarrow \log_3 9 \Rightarrow k$

$(3)^k \Leftrightarrow 9 \quad \{3^k \Rightarrow 3^2\}$

$k \Rightarrow 2$

$\rightarrow \log_2 \log_2 2$

$\rightarrow \log_2 2 \Rightarrow 1 \quad \{\log_a a \Rightarrow 1\}$

$\Rightarrow \log_2 1 \Rightarrow \underline{0}$

$\therefore \log_2 \log_2 \log_3 \log_3 27^3 \Rightarrow 0$

$x \Rightarrow \sqrt{5\sqrt{5\sqrt{5}\ldots\infty}}$

$\{\because O.B.\}$

$x^2 \Rightarrow 5\sqrt{5\sqrt{5}\ldots\infty}$

$x^2 \Rightarrow 5\sqrt{5\sqrt{5}\ldots\infty}$

$x \Rightarrow \sqrt{5\sqrt{5\sqrt{5}\ldots\infty}}$

Divide x^4 by x

$x \neq 0$

$x \Rightarrow 5$

(or)

$x^2 \Rightarrow 5x$

$x^2 - 5x$

Ex) $\log_5 \sqrt{5\sqrt{5\sqrt{5}}}\ldots\infty$

$\Rightarrow \log_5 \sqrt{5\sqrt{5\sqrt{5}}}\ldots\infty \Rightarrow k$

$(5)^k \Rightarrow \sqrt{5\sqrt{5\sqrt{5}}}\ldots\ldots\infty$

$(5)^k \Rightarrow 5$

$(5)^k \Leftrightarrow (5)^1$ ∴

$k \Rightarrow 1$

$\Rightarrow \log_5 \sqrt{5\sqrt{5\sqrt{5}}}\ldots\infty \Rightarrow \underline{1}$

$x \Rightarrow \sqrt{5\sqrt{5\sqrt{5}}}\ldots\infty$

$x^2 \Rightarrow 5\sqrt{5\sqrt{5}}\ldots\infty$

$x^2 \Rightarrow 5\sqrt{5\sqrt{5}}\ldots\infty$

$x^2 \Rightarrow 5\sqrt{5}\sqrt{5}\ldots\infty$

Divide

$\frac{x^2}{x} \Rightarrow \frac{5\sqrt{5\sqrt{5}}\ldots\infty}{\sqrt{5\sqrt{5\sqrt{5}}}\ldots\infty}$

$x \neq 0 \{\frac{1}{q}, q \neq 0\}$

$\underline{x \Rightarrow 5}$

Note:

1) $\log_a x \Rightarrow$
   i) $x > 0$
   ii) $a > 0$
   iii) $a \neq 1$

$\Rightarrow \log_3 -2 \Rightarrow$ not. defined

$\log_{-5} 9 \Rightarrow$ not defined

$\log_1 100 \Rightarrow$ not defined

In Log It has 2 famous Base

⇒ ∴ 2 Base ⇔ World famous

1) → $\log_{10} x$ → commonly used base in Log

$\log_e x$ → Natural Log

— natural Base

$\pi \approx 3.1415...$

$e \approx 2.71828...$

Euler's constant / napier's Constant

∴ $\ln x \iff \log_e x$

Properties of Log

1) $\log_a m + \log_a n \Leftrightarrow \log_a m.n$

2) $\log_a m - \log_a n \Leftrightarrow \log_a m/n$

3) $\log_c a^b \Leftrightarrow b \log_c a$

4) $\log_a b \Leftrightarrow \frac{\log_c b}{\log_c a}$ {∵ base should be same}

$\Leftrightarrow \frac{1}{\log_b a}$

5) $a^{\log_a N} \Leftrightarrow N$

{e.g $\Rightarrow 7^{\log_7 5} \Rightarrow 5^{\log_7 7} \Rightarrow 5^1 \Rightarrow 5$}

6) $\log_{c^d} b^a \Leftrightarrow \frac{a}{d} \log_c b$

7) $a^{\log_b c} \Leftrightarrow c^{\log_b a}$

Ex) $\log_a b \propto \log_b a \Rightarrow ?$

$\rightarrow \log_a b \times \log_b a$

$\rightarrow \frac{\log b}{\log a} \times \frac{\log a}{\log b}$

$\rightarrow 1$

$\Rightarrow \log_a b \times \log_b a \Rightarrow \underline{1}$

Ex) $\log_a b \log_b c \log_c d \log_d e \cdot \log_e a \Rightarrow ?$

$\rightarrow \frac{\log b}{\log a} \times \frac{\log c}{\log b} \times \frac{\log d}{\log c} \times \frac{\log e}{\log d} \times \frac{\log a}{\log e}$ $\left\{ \log_a b \Rightarrow \frac{\log b}{\log a} \right\}$

$\rightarrow \underline{1}$

Ex) $\log_2 3 \log_3 4 \log_4 5 \ldots \log_n n+1 \Rightarrow 10$ find $n \Rightarrow ?$

$\rightarrow \log_2 3 \log_3 4 \log_4 5 \ldots \log_n n+1 \Rightarrow 10$

$\rightarrow \frac{\log 3}{\log 2} \frac{\log 4}{\log 3} \frac{\log 5}{\log 4} \ldots \frac{\log n+1}{\log n} \Rightarrow 10$

$\rightarrow \frac{\log n+1}{\log 2} \Rightarrow 10$

$\rightarrow \frac{\log n}{\log 2} \times \frac{\log 1}{\log n} \times \frac{\log 1}{\log 2} \Rightarrow 10$

$\rightarrow \frac{\log 1}{\log 2} \times \frac{1}{\log 2} \Rightarrow 10$

→ $\log_n 1 \times \frac{1}{\log 2} \Rightarrow 10$

→ $\log 1 \times \frac{1}{\log 2} \Rightarrow n^{10}$

$\Rightarrow \frac{\log 3}{\log 2} \log_3 4 \log_4 5 \cdots \log_n n+1 \Leftrightarrow 10$

→ $\frac{\log 3}{\log 2} \frac{\log 4}{\log 3} \frac{\log 5}{\log 4} \cdots \frac{\log n+1}{\log n} \Leftrightarrow 10$

→ $\frac{\log n+1}{\log 2} \Leftrightarrow 10$

→ $\log_2 n+1 \Leftrightarrow 10$

→ $\log_2 n+1 \Leftrightarrow 10$

$\Rightarrow n+1 \Rightarrow 2^{10}$

$\Rightarrow n \Rightarrow 2^{10} - 1$

$\log_2 2^{10} - 1 \Rightarrow 10$

$n \Rightarrow \underline{1023}$

Ex) $A \Rightarrow \log_{11} 11^{(\log_{11} 1331)}$

$B \Rightarrow \log_{385} 5 + \log_{385} 7 + \log_{385} 11$

$C \Rightarrow \log_4(\log_2(\log_5 625))$

$D \Rightarrow 10^{\log_{100} 16}$ find $\frac{AB}{CD}$

$\Rightarrow$

$\therefore A \Rightarrow \log_{11} 11^{(\log_{11} 1331)}$

$\Rightarrow \log_{11} 1331 (\log_{11} 11) \Rightarrow 3$

$1 \Rightarrow \log_a a \Rightarrow 1$

$\log_a b^c \Rightarrow c \log_a b$

$\Rightarrow \log_{11} 1331 \Rightarrow k$

$\Rightarrow 11^k \Rightarrow 1331$

$\Rightarrow 1.1^k \Rightarrow 11^3$

$k \Rightarrow 3$

$\rightarrow \log_{11} 1331 \Rightarrow 3$

$\Rightarrow \log_{11} 11^{\log_{11} 1331} \Rightarrow 3$

$\therefore B \Rightarrow \log_{385} 5 + \log_{385} 7 + \log_{385} 11 \Rightarrow 1$

$\rightarrow \{ \log_a m + \log_a n + \log_a q \Rightarrow \log_a m \cdot n \cdot q \}$

$\Rightarrow \log_{385} 5 \times 7 \times 11$

$\Rightarrow \log_{385} 385 \Rightarrow 1$

$\therefore C \Rightarrow \log_4 (\log_2 (\log_5 625)) \Rightarrow \frac{1}{2}$

$\log_4 (\log_2 (\log_5 625))$

$\log_5 625 \Rightarrow k$

$5^k \Rightarrow 625$

$5^k \Rightarrow 5^4$

$k \Rightarrow 4$

$\log_4 \log_2 4$

$\log_4 2$

$k \Rightarrow \frac{1}{2}$

$\therefore$ find $\frac{AB}{CD} \Rightarrow \frac{A \times B}{C \times D} \Rightarrow \frac{3 \times 1}{\frac{1}{2} \times 4}$

$\Rightarrow \frac{3}{2}$

$\Rightarrow 10^{\log_{100} 16}$

$\{ \log_b c \Rightarrow c^{\log_b a} \} \Rightarrow 16^{\log_{100} 10} \cdot \{ \log_{100} 10 \Rightarrow \frac{1}{2} \}$

$\Rightarrow 4^{\left(2 \times \frac{1}{2}\right)}$

$\Rightarrow \underline{\underline{4}}$

{Co-prime → Two numbers are co-prime (or relatively prime) if their only common factor is "1"

Ex) e.g. 8 and 15

If $N \Rightarrow \frac{\log_3 12}{\log_{36} 3} - \frac{\log_3 4}{\log_{108} 3}$

8 ⇒ 1, 2, 4, 8
15 ⇒ 1, 3, 5, 15
Common factor 1

6 and 9
6 ⇒ 1, 2, 3, 6
9 ⇒ 1, 3, 9
Common factor ⇒ 1, 3
(no coprime)

the N is (i) Composite No.

(ii) ~~Prime No.~~

(iii) ~~Co-prime With 7~~

(iv) Rational number which is not an Integer.

=)

$N \Rightarrow \frac{\log_3 12}{\log_{36} 3} = \frac{\log_3 4}{\log_{108} 3}$

$\Rightarrow \frac{\log 12 \times \log 36}{\log 3 \times \log 3} - \frac{\log 4 \times \log 108}{\log 3 \times \log 3}$

Assume all log base is 10.

$\frac{\log 12 \times \log 36}{\log 3 \times \log 3} - \frac{\log 4 \times \log 108}{\log 3 \times \log 3}$

$\frac{\log_{10} 12+36}{\log_{10} 3+3} - \frac{\log_{10} 4+108}{\log_{10} 3+3}$

$$\rightarrow \frac{\log_{10} 18}{\log_{10} 6} = \frac{\log_{10} 112}{\log_{10} 6}$$

$$\rightarrow \frac{\log_{10} \frac{\cancel{18}}{\cancel{112}_{56}}}{\log_{10} 6} \Rightarrow \frac{\log_{10} \frac{6}{19}}{\log_{10} 6} \Rightarrow \log_{19} 6$$

$$\Rightarrow$$

$$N \Rightarrow \frac{\log_3 12}{\log_{36} 3} - \frac{\log_3 4}{\log_{108} 3}$$

$$\rightarrow (\log_3 12)[\log_{3} 36] - (\log_3 4)(\log_3 108)$$

$$\rightarrow \left\{ \log_a b \Rightarrow \frac{1}{\log_b a} \right\}$$

=) Prime factor the following eq^n

$$=) (\log_3 2^2 \cdot 3)(\log_3 2^2 \cdot 3^2) - (\log_3 2^2)(\log_3$$

$$\Rightarrow \{\log_a mn \Rightarrow \log_a m + \log_a n\}$$

$$(\log_3 2^2 + \log_3 3)(\log_3 2^2 + \log_3 3^2) - (\log_3 2^2)($$

$$(\log_a m^n \Rightarrow n \log_a m)$$

$$2\log_3 2 + 1][2\log_3 2 + 2(1)] - [2\log_3 2][3($$

Assume f_n $\Rightarrow f$

$$= \frac{(2t+1)(2t+2) - (2t)(3+2t)}{}$$

$$= 4t^2 + 4t + 2t + 2 - 6t - 4t^2$$

$$= 2$$

Logarithmic Equation

Ex) Find the value of x?

$\Rightarrow x^2 + 1^{\log_7 x} - 2 = 0$

$\Rightarrow x^2 + 7^{\log_7 x} - 2 \Rightarrow 0$

$\Rightarrow x^2 + x^{\log_7 7} - 2 \Rightarrow 0$ ($\log_7 7 \to 1$)

$\Rightarrow x^2 + x - 2 \Rightarrow 0$

$\Rightarrow x^2 + 2x - x - 2 \Rightarrow 0$ ($2x^2$: $2x$, x)

$\Rightarrow x(x+2) - 1(x+2)$

$\Rightarrow (x+2)(x-1)$

$x \Rightarrow -2$

$x \Rightarrow 1$

$x > 0$ → Logrithic Condition

$\therefore x \Rightarrow 1$

$(1)^2 + 1 - 2 \Rightarrow 0$

$2 - 2 \Rightarrow 0$

Ex) Sum of all the Solution(s) of the Equation

$\log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) \Rightarrow 0$

$\Rightarrow \log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) \Rightarrow 0$

$\Rightarrow \log_{10}(x)(x+2) \{\log_a m + \log_a n \Rightarrow \log mn\} - \log_{10}(5x+4) \Rightarrow 0$

$\Rightarrow \log_{10}(x^2+2) - \log_{10}(5x+4) \Rightarrow 0$

$\Rightarrow \log_{10}\frac{(x^2+2x)}{5x+4} \Rightarrow 0 \quad \{\log_a m - \log_a n \Rightarrow \log \frac{m}{n}\}$

$\Rightarrow \frac{(x^2+2x)}{(5x+4)} \Rightarrow (10)^0$

$\Rightarrow x^2+2x \Rightarrow (5x+4)$

$\Rightarrow x^2+2x-5x-4 \Rightarrow 0$

$\Rightarrow x^2-5x-4+2x \Rightarrow 0$

$\Rightarrow x^2-3x-4 \Rightarrow 0$

$4x^2$

$\rightarrow x^2 - 4x + x - 4 \Rightarrow 0 \quad 4x \quad x$

$\rightarrow x(x-4)+1(x-4) \Rightarrow 0$

$\Rightarrow (x-4)(x+1) \Rightarrow 0$

$\therefore x \Rightarrow 4$ and $x \Rightarrow -1$